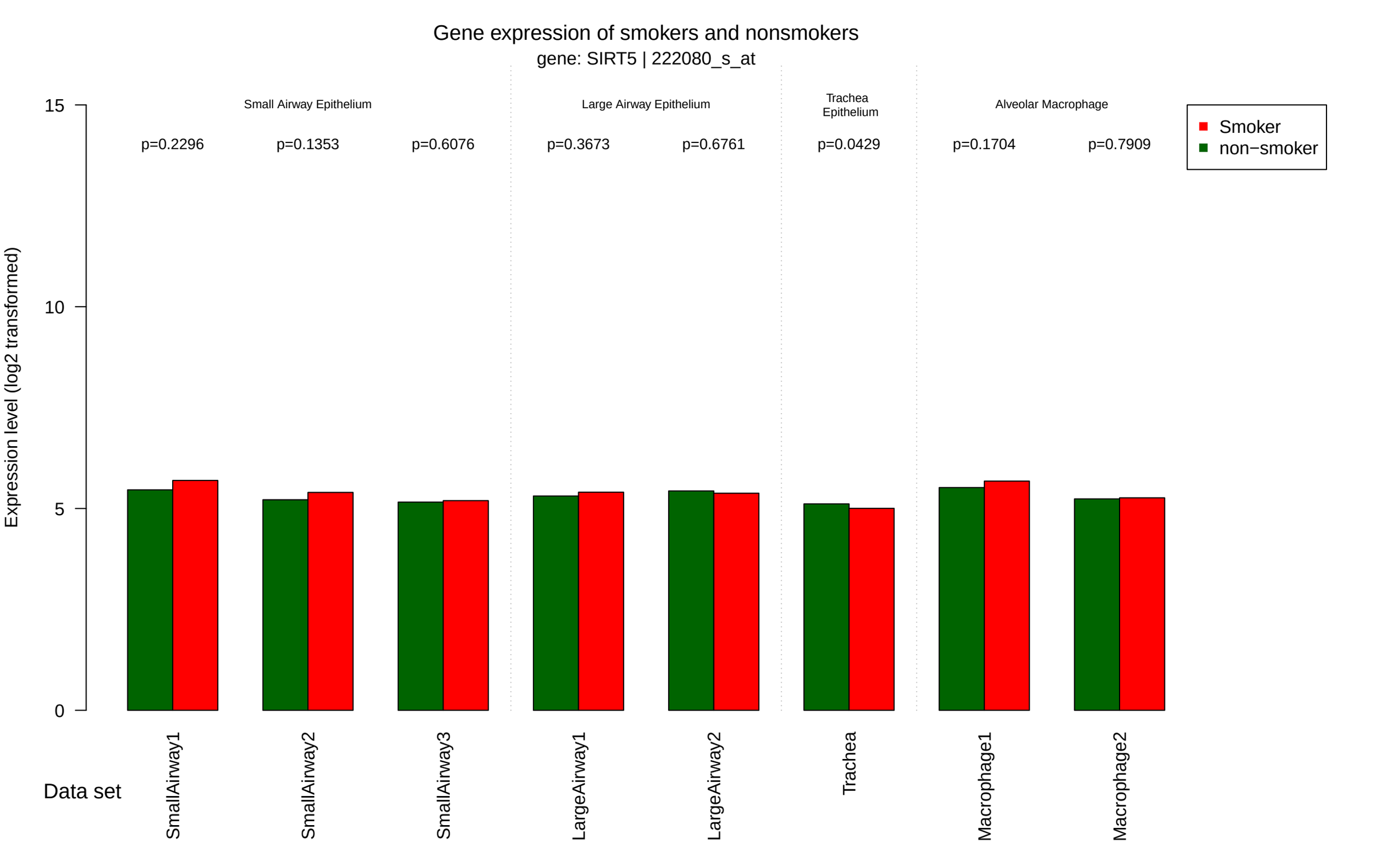
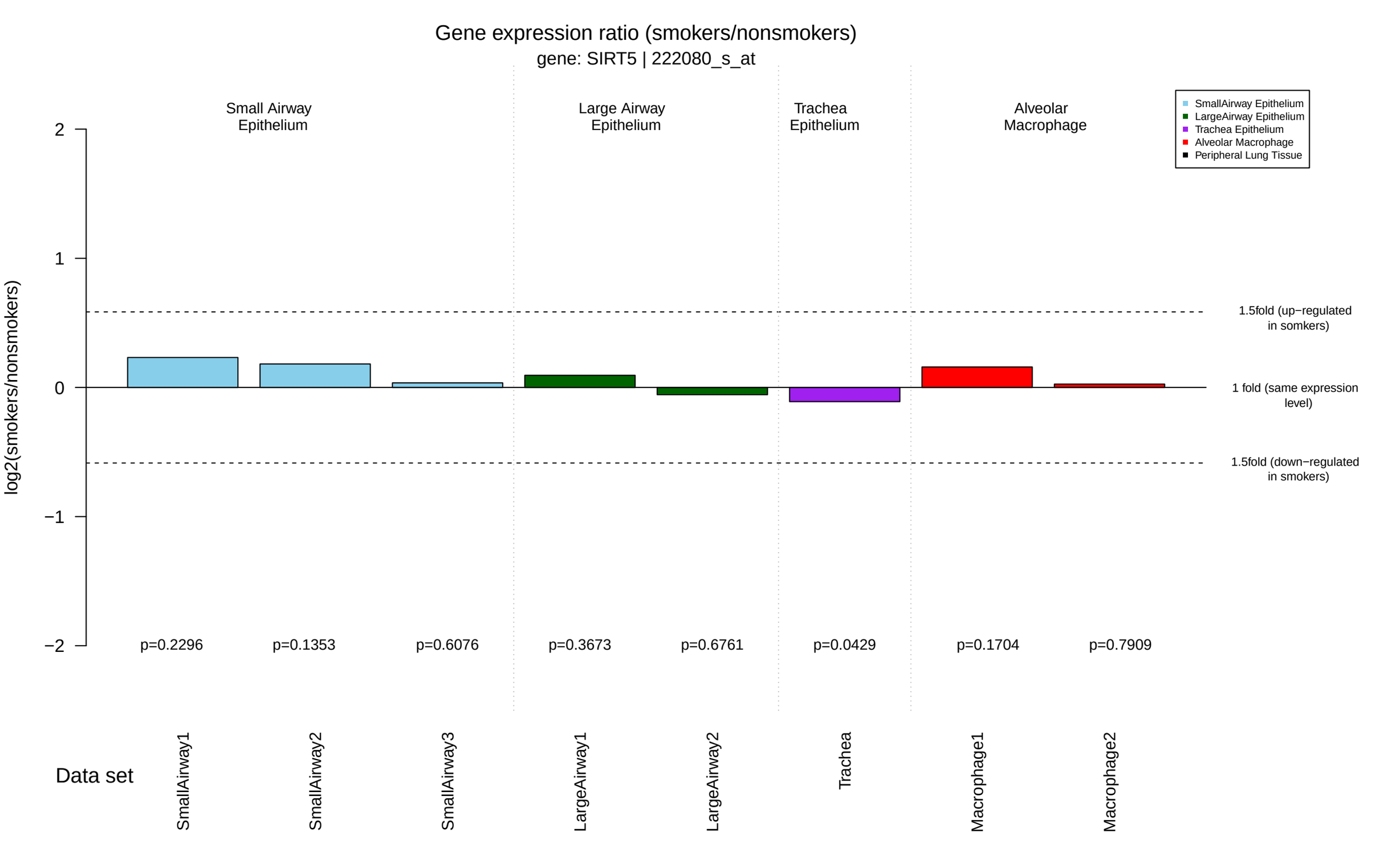
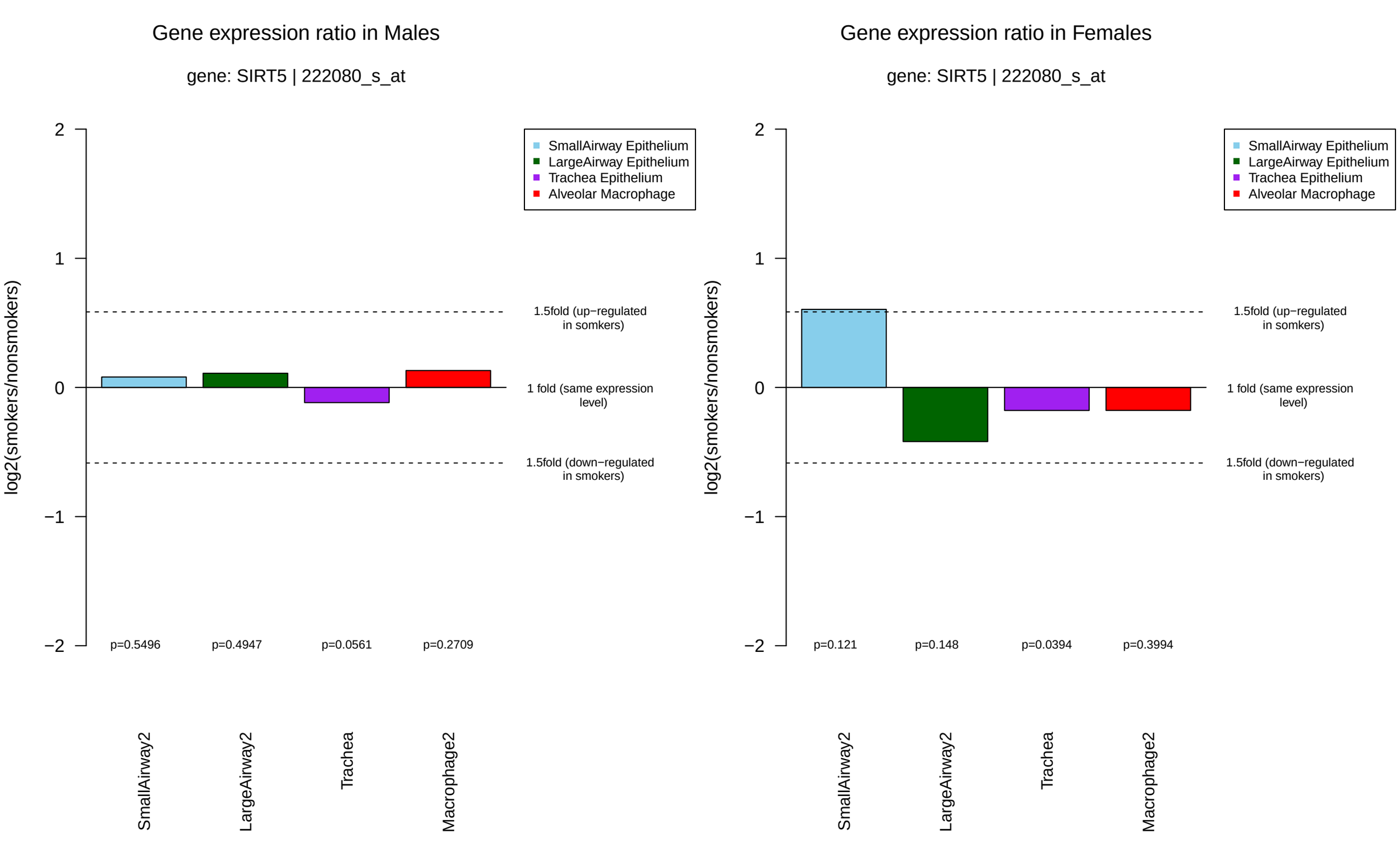
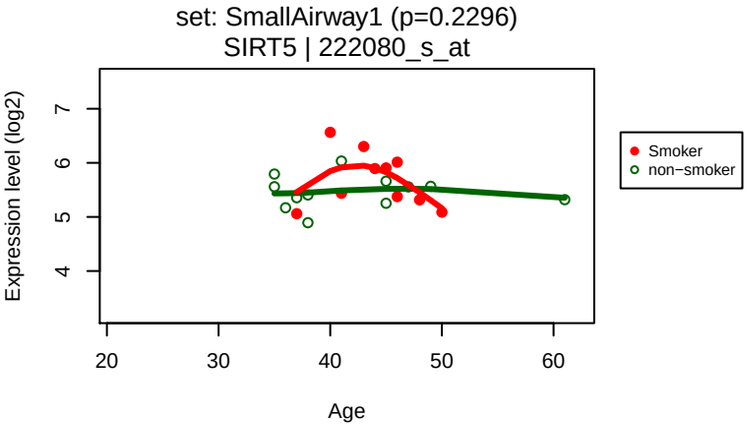
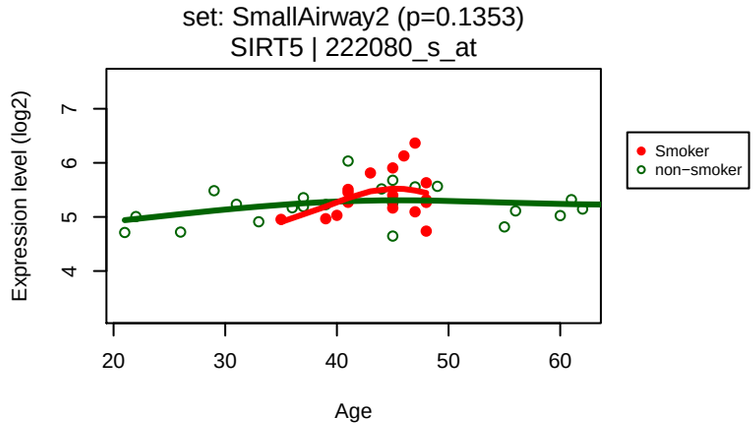
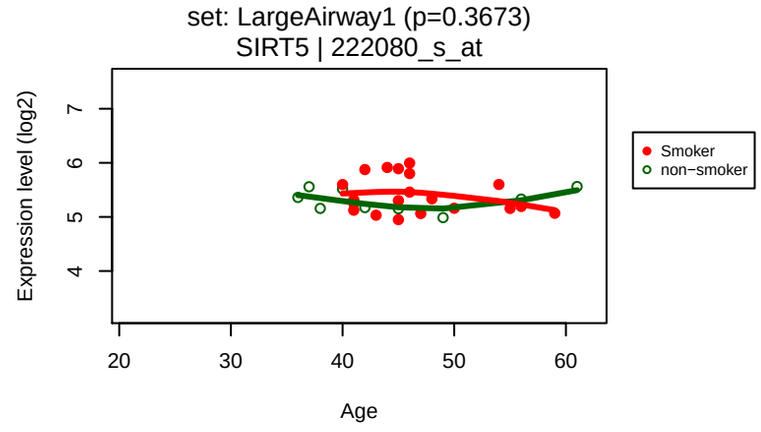

Gene expression of smokers and nonsmokers
gene: SIRT5 | 222080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2296
p=0.1353
p=0.6076
p=0.3673
p=0.6761
p=0.0429
p=0.1704
p=0.7909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

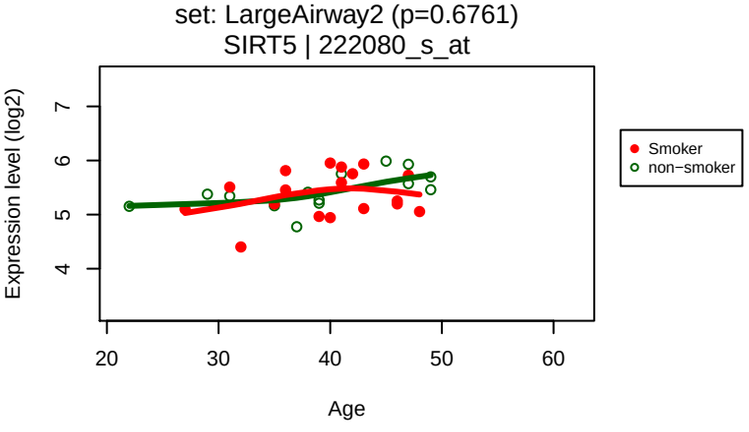

Gene expression ratio (smokers/nonsmokers)
gene: SIRT5 | 222080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2296
p=0.1353
p=0.6076
p=0.3673
p=0.6761
p=0.0429
p=0.1704
p=0.7909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

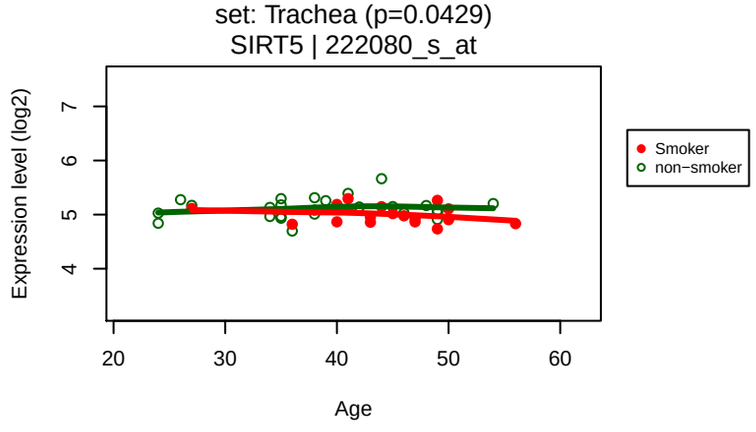

Gene expression ratio in Males
gene: SIRT5 | 222080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5496
p=0.4947
p=0.0561
p=0.2709
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT5 | 222080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.121
p=0.148
p=0.0394
p=0.3994
SmallAirway2
LargeAirway2
Trachea
Macrophage2

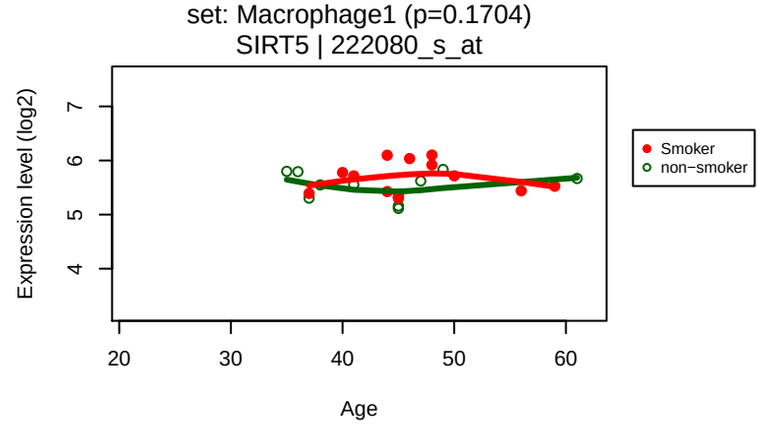

set: SmallAirway1 (p=0.2296)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker

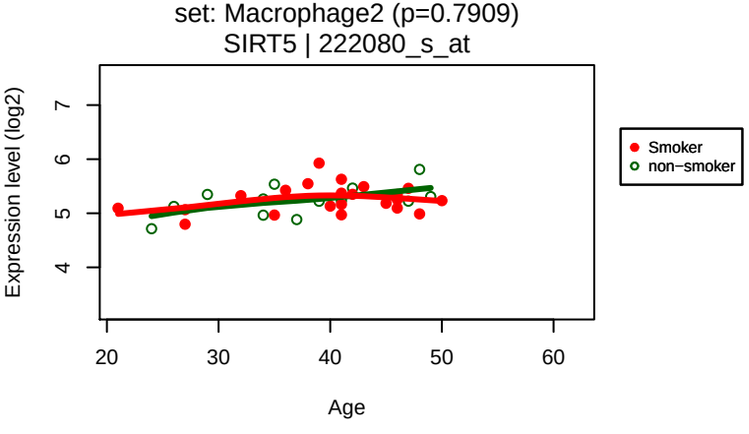

set: SmallAirway2 (p=0.1353)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker

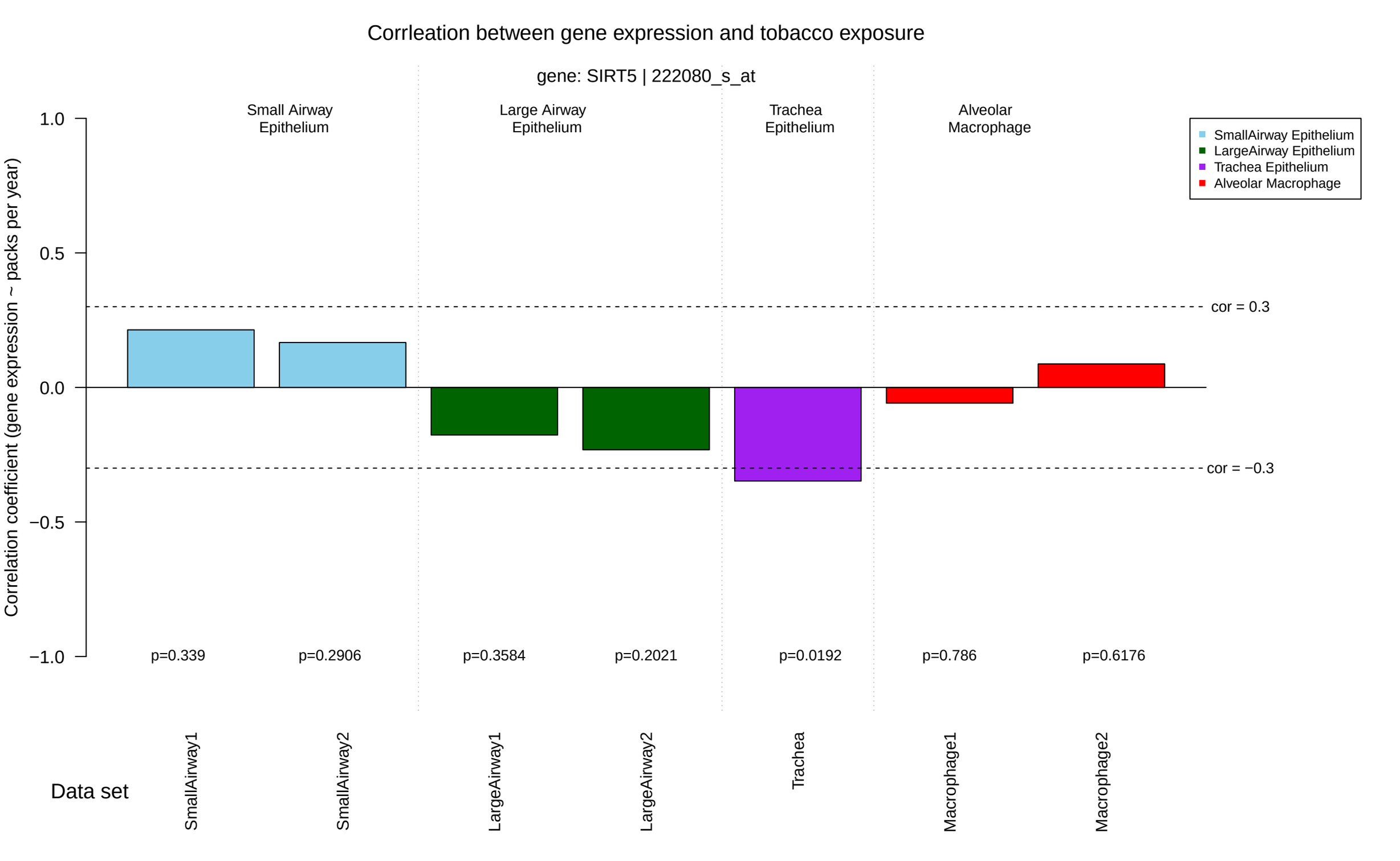

set: LargeAirway1 (p=0.3673)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6761)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0429)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1704)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7909)
SIRT5 | 222080_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT5 | 222080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.339
p=0.2906
p=0.3584
p=0.2021
p=0.0192
p=0.786
p=0.6176
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2